

FinRelief AI — API Documentation

SmartBridge Internship Program

Version 2.0.0

1. Base URL and Conventions

All endpoints are served under the FastAPI backend and return JSON. Interactive Swagger documentation is auto-generated by FastAPI and available at /docs on the running backend instance.

Every request or response body is validated by a Pydantic schema. Requests with missing or incorrectly typed fields are rejected before any business logic runs, with a descriptive validation error.

2. Authentication

Authentication uses stateless JSON Web Tokens. On successful login or registration, the backend issues a signed JWT containing the user's ID and an expiry. The frontend attaches this token to every subsequent request:

Authorization: Bearer <token>

Passwords are never stored in plain text — they are hashed with bcrypt before being written to the database. Tokens are signed with a secret key loaded from the environment (FINRELIEF_SECRET_KEY), and the backend verifies the signature on every protected request rather than maintaining server-side session state.

Endpoint	Method	Auth Required	Description
/auth/register	POST	No	Registers a new user and returns a JWT
/auth/login	POST	No	Authenticates an existing user and returns a JWT
/auth/me	GET	Yes	Returns the currently authenticated user's details
/auth/logout	POST	Yes	Blacklists the current JWT

3. Loans

Full CRUD for a user's loans. All endpoints in this group require authentication and only operate on loans owned by the requesting user.

Endpoint	Method	Description
/loans	GET	Lists all loans belonging to the authenticated user
/loans	POST	Creates a new loan record
/loans/{id}	GET	Retrieves a single loan by ID
/loans/{id}	PATCH	Updates fields on an existing loan
/loans/{id}	DELETE	Deletes a loan

A loan record captures the lender name, loan amount, EMI, number of days overdue, and the borrower's monthly income — the four inputs the settlement calculation is built from.

4. Settlement

Endpoint	Method	Description
/settlement/{loan_id}	POST	Runs the settlement calculation for a loan and saves a stress snapshot
/snapshots	GET	Returns all stress snapshots across the user's loans, used for dashboard trend charts
/snapshots/{loan_id}	GET	Returns snapshots for one specific loan

Running a settlement calculation computes the debt-to-income ratio, a debt stress level, and a recommended settlement percentage from the loan's stored financial data, and stores the result as a snapshot so the dashboard can chart it over time.

5. Letters

Endpoint	Method	Description
/letters/{loan_id}	POST	Generates an OTS negotiation letter for a loan, via Gemini or the fallback template
/letters	GET	Returns the latest letter for each of the user's loans

Endpoint	Method	Description
/letters/{loan_id}	GET	Returns the latest letter for one specific loan
/letters/{loan_id}/history	GET	Returns every letter version generated for a loan
/letters/download/{letter_id}	GET	Streams a letter back as a downloadable PDF
/letters/{letter_id}	PUT	Edits the body of a saved letter

When a letter is generated, the backend first attempts to call Google Gemini 2.0 Flash to draft the letter body from the loan's financial profile. If the Gemini call succeeds, the letter is saved and tagged as AI-generated. If it fails for any reason — timeout, a non-2xx response, or a response that can't be parsed — the backend falls back to a static, still-professional template automatically, and the letter is tagged accordingly. Either way, the user receives a usable letter and the source is recorded honestly rather than silently.

Letters can be downloaded as PDFs, generated through ReportLab, so a user can save or print the letter for their own records or to send directly.

6. Dashboard

Endpoint	Method	Description
/dashboard	GET	Returns aggregated statistics across all of the user's loans, snapshots, and letters for the dashboard view

7. Demo and Health

Endpoint	Method	Description
/demo/seed	POST	Seeds three sample loans into a demo account, so a new visitor can explore the app without entering real financial data
/health	GET	Reports database connectivity and Gemini API availability, used for monitoring

8. Rate Limiting

Sensitive endpoints — particularly authentication and letter generation — are rate-limited using SlowAPI to reduce the risk of abuse, such as repeated login attempts or excessive AI calls from a single client.

9. Error Handling

Validation errors return a 422 response with a description of which field failed and why, generated automatically by Pydantic. Authentication failures return 401. Attempting to access or modify a resource that doesn't belong to the requesting user returns 404 rather than 403, to avoid confirming the existence of another user's data.